Predicting Urban Safety Perception from Street-View Images: A Comparative Study of CNN and Vision Transformer

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Abstract

This study investigates the effectiveness of deep learning models in predicting urban safety perception from street-view images using the MIT Place Pulse 2.0 dataset. We conduct a systematic comparison of two preprocessing methods (RGB vs. Grayscale) and two model architectures (ResNet50 vs. Vision Transformer) to understand how color information and architectural choices affect prediction accuracy. Our experiments reveal that RGB preprocessing achieves marginally higher average accuracy (72.10%) compared to Grayscale (69.83%), with a color contribution of +2.28%. Interestingly, ResNet50 slightly outperforms ViT on average (71.61% vs. 70.32%), contrary to recent trends favoring Transformers. Most notably, we discover a significant interaction effect: while ViT benefits substantially from color information, ResNet50 shows robust performance regardless of color presence, suggesting fundamentally different feature extraction strategies between these architectures.

1. Introduction

Urban safety perception plays a crucial role in urban planning, real estate valuation, and public policy decisions. Understanding how people perceive safety from visual cues in street environments can inform evidence-based interventions to improve urban livability. Traditional methods of assessing neighborhood safety rely on crime statistics or surveys, which may not capture the nuanced visual features that influence human perception.

The MIT Place Pulse 2.0 dataset [1] provides a unique opportunity to study this problem through crowdsourced human judgments. The dataset contains over 110,000 street-view images from 56 cities worldwide, with pairwise comparison votes indicating which location appears "safer." These votes are aggregated using the TrueSkill algorithm to produce continuous safety scores for each location.

This study addresses two key research questions: (1) How much does color information contribute to safety perception prediction? Prior work suggests structural features like building density and street width may dominate [2], yet color cues such as greenery and lighting conditions could also be significant. (2) Do Vision Transformers, which have shown remarkable success in various computer vision tasks, outperform traditional CNNs for this perceptual judgment task?

Dataset Description: We processed the Place Pulse 2.0 dataset to create a binary classification task. Using a threshold of mean plus/minus 1.5 standard deviations on TrueSkill scores, we identified 7 distinct places with clear safety labels, which expanded to 11,296 images through the location mapping. The final dataset contains 47.8% unsafe and 52.2% safe labels, providing a well-balanced classification problem. We employed place-based data splitting to ensure no location overlap between training and validation sets, preventing data leakage from location-specific features.

2. Related Work

Dubey et al. [1] introduced the Place Pulse 2.0 dataset and established baseline performance using a Siamese-style CNN architecture, achieving 73.5% accuracy on pairwise safety comparisons. Their work demonstrated that deep features could capture human safety perception effectively.

Naik et al. [2] explored the relationship between visual appearance and perceived safety, finding that structural elements such as visible sky, greenery, and building facades correlate strongly with safety perception. This work suggests that grayscale features might capture substantial predictive information.

Recent advances in Vision Transformers [3] have shown superior performance on various image classification benchmarks. Zhang et al. [4] combined semantic segmentation with ViT architectures, achieving 91.7% on urban perception tasks. However, the relative contribution of color versus structural features, and the comparative effectiveness of CNN versus Transformer architectures on this specific task, remains underexplored.

3. Methods

3.1 Data Preprocessing

We implemented two preprocessing pipelines to investigate the role of color information. **RGB preprocessing** maintains all three color channels, preserving information about greenery, building materials, sky conditions, and lighting. **Grayscale preprocessing** converts images to single-channel luminance (replicated to three channels for model compatibility), forcing the model to rely solely on structural features such as edges, textures, and spatial arrangements.

3.2 Data Augmentation

We applied horizontal flipping (p=0.5) and color jitter (brightness=0.2, contrast=0.2) during training. Vertical flipping and rotation were deliberately excluded as they would create unrealistic urban perspectives. All images were resized to 224x224 pixels and normalized using ImageNet statistics.

3.3 Model Architectures

ResNet50 [5] serves as our CNN baseline. This 50-layer residual network (23.5M parameters) uses skip connections to enable training of deep networks. We replaced the final fully-connected layer with a binary classification head and initialized from ImageNet-pretrained weights.

Vision Transformer (ViT-B/16) [3] represents the Transformer-based approach. This model (86M parameters) divides input images into 16x16 patches, projects them linearly, and processes them through 12 Transformer encoder layers with global self-attention. The [CLS] token representation feeds into a classification head.

3.4 Training Configuration

All models were trained for 15 epochs using AdamW optimizer (lr=1e-4, weight_decay=0.01) with cosine annealing learning rate scheduling. We used batch size 32 and cross-entropy loss. Place-based splitting ensured 80% of locations (5 places) for training and 20% (2 places) for validation, with zero location overlap to evaluate true generalization capability. All experiments were conducted on an Apple M4 GPU with MPS (Metal Performance Shaders) acceleration, completing in approximately 2.5 hours total.

4. Results

4.1 Error Analysis

Figure 1 shows the confusion matrices for all four configurations. The RGB+ViT model (Figure 1(b)) achieved 73.23% accuracy with **286 false positives** (predicting safe when actually unsafe) and **961 false negatives**.

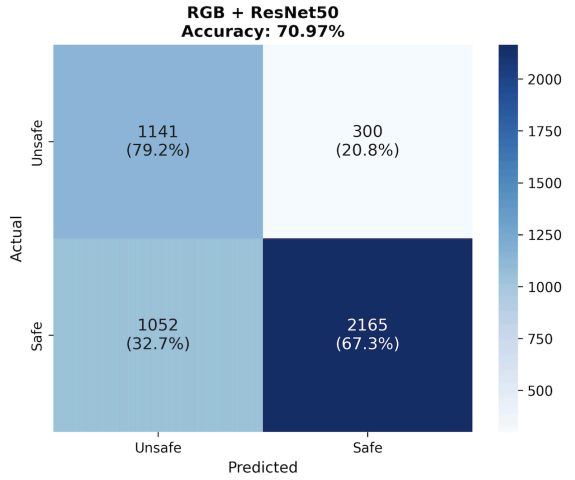


Figure 1(a): *Confusion matrix for RGB+ResNet50*

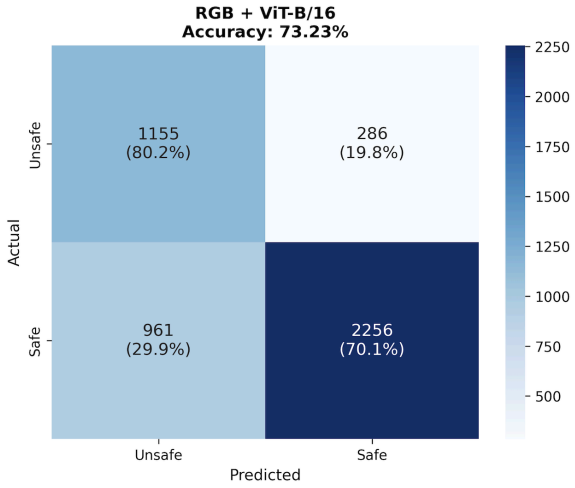


Figure 1(b): *Confusion matrix for RGB+ViT-B/16*

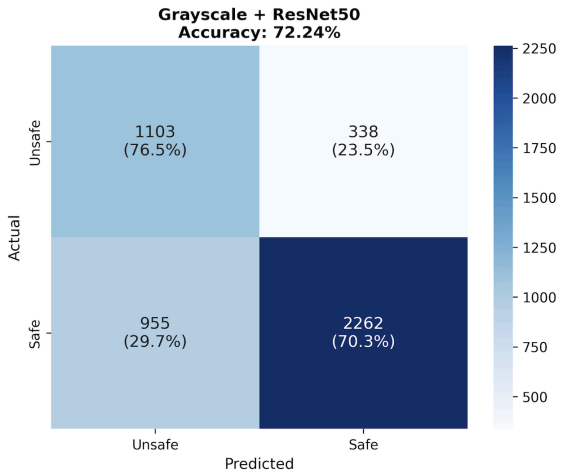


Figure 1(c): *Confusion matrix for Grayscale+ResNet50*

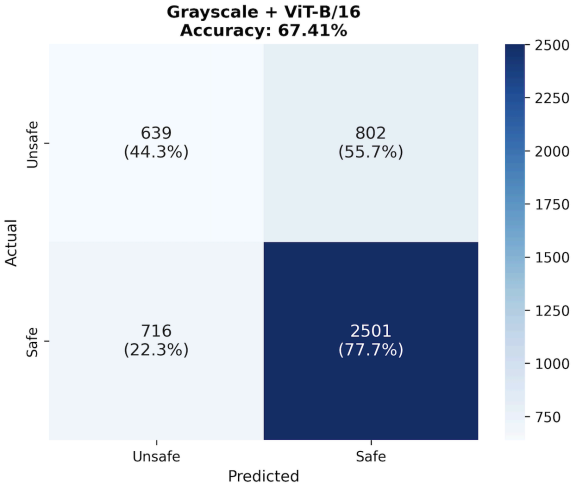


Figure 1(d): *Confusion matrix for Grayscale+ViT-B/16*

Figure 2 displays representative misclassified examples from the RGB+ViT model. **False positives** (figure 2(a) and figure 2(b) predicted safe but actually unsafe) often contain visually ambiguous cues such as flat, straight roads and orderly parked vehicles—features typically associated with well-maintained neighborhoods, which may have misled the model.



Figure 2(a): False positive examples (predicted safe, actually unsafe)



Figure 2(b): False positive examples (predicted safe, actually unsafe)

False negatives (Figure 2(c) and 2(d)) predicted unsafe but actually safe) tend to feature large vehicles (e.g., trucks or buses) occupying significant portions of the frame and low-rise buildings, suggesting the model may over-associate vehicle size and building height with perceived danger.



Figure 2(c): False negative examples (predicted unsafe, actually safe)



Figure 2(d): False negative examples (predicted unsafe, actually safe)

Table 1 summarizes the validation accuracy across all experimental configurations. The best overall performance (73.23%) was achieved by RGB + ViT, while the lowest (67.41%) came from Grayscale + ViT.

Configuration	Val Accuracy
RGB + ResNet50	70.97%
RGB + ViT-B/16	**73.23%**
Grayscale + ResNet50	72.24%
Grayscale + ViT-B/16	67.41%

Table 1: *Validation Accuracy Across All Configurations*

Key Findings:

Color Contribution: RGB preprocessing achieves an average accuracy of 72.10% compared to 69.83% for Grayscale, yielding a modest +2.28% improvement. This suggests that while color provides useful information, structural features capture the majority of safety-relevant visual cues.

Architecture Comparison: Contrary to expectations, ResNet50 (71.61% average) slightly outperforms ViT-B/16 (70.32% average) by 1.29%. This may be attributed to the relatively small dataset size (11,296 images), where CNNs' inductive biases provide advantages over Transformers' data-hungry nature.

Interaction Effect: The most striking finding is the significant interaction between preprocessing and architecture. ResNet50 shows remarkable robustness to color removal, actually improving from 70.97% to 72.24% with Grayscale. In contrast, ViT experiences a dramatic 5.82% drop (73.23% to 67.41%) when color is removed, indicating its reliance on color-based attention patterns.

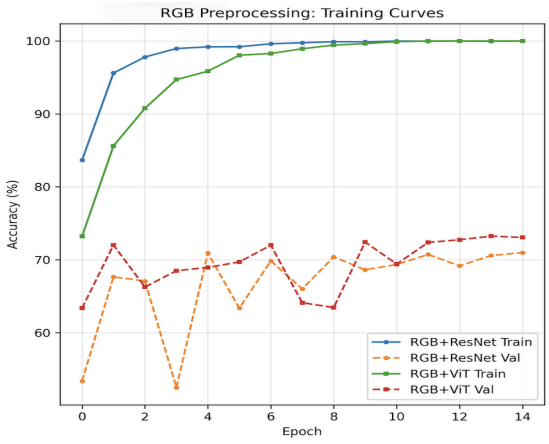


Figure 3: *Training and validation accuracy curves for RGB preprocessing across ResNet50 and ViT-B/16 architectures.*

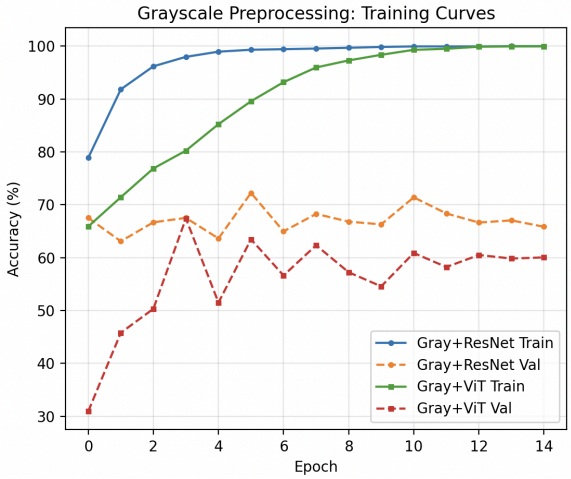


Figure 4: *Training and validation accuracy curves for Grayscale preprocessing across ResNet50 and ViT-B/16 architectures.*

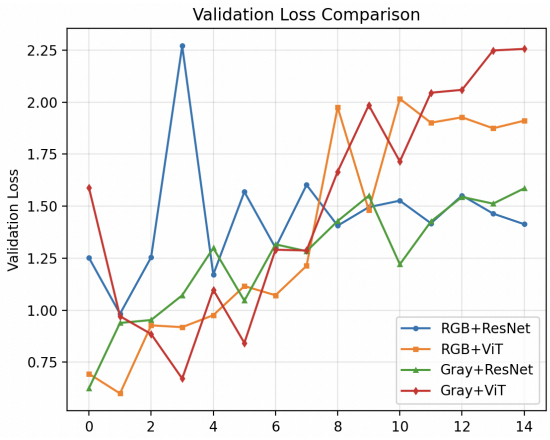


Figure 5: *Validation loss comparison across all four experimental configurations over 15 epochs.*

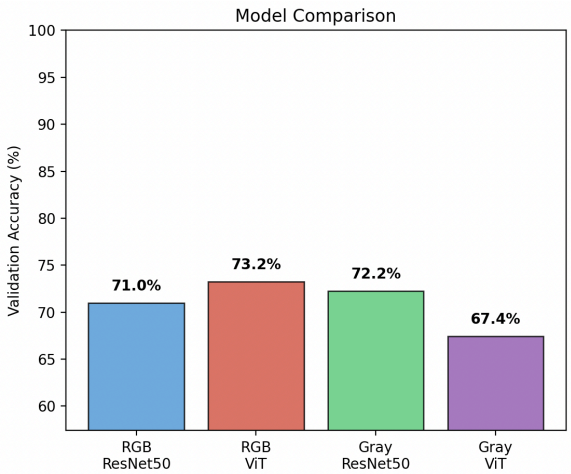


Figure 6: *Validation accuracy comparison across all model configurations. RGB+ViT achieves the highest*

accuracy (73.2%).

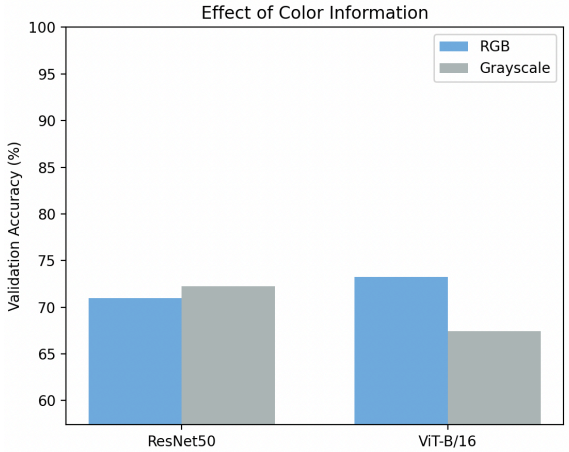


Figure 7: *Effect of color information on model performance. ViT shows greater sensitivity to color removal compared to ResNet50.*

Configuration	Accuracy
RGB + ResNet50	70.97%
RGB + ViT	73.23%
Grayscale + ResNet50	72.24%
Grayscale + ViT	67.41%
RGB Average	72.10%
Grayscale Average	69.83%
Color Contribution	+2.28%
ResNet50 Average	71.61%
ViT Average	70.32%
Architecture Gain	-1.29%

Table 2: *Summary of key experimental insights including color contribution (+2.28%) and architecture comparison (-1.29%).*

5. Discussion

Our results provide several insights into urban perception modeling. The modest color contribution (+2.28%) aligns with prior work suggesting that structural features dominate safety perception [2]. However, the architecture-dependent nature of this effect reveals that different model families extract fundamentally different feature representations.

The unexpected finding that Grayscale+ResNet50 outperforms RGB+ResNet50 warrants further investigation. One hypothesis is that color variation across different cities and lighting conditions introduces distribution shift that harms generalization. ResNet50's convolutional structure may be better suited to learning location-invariant structural patterns when color is removed.

Limitations: (1) Our dataset, while containing over 11,000 images, derives from only 7 distinct places after filtering, limiting geographic diversity. (2) Binary classification simplifies the continuous nature of safety perception. (3) We did not explore semantic segmentation or attention visualization, which could provide interpretability. (4) The place-based split, while methodologically sound, may result in high variance due to the small number of validation locations.

Future Work: Several directions could extend this study: (1) incorporating semantic segmentation to identify which visual elements (sky, greenery, buildings) contribute most to safety perception, (2) attention visualization to interpret ViT's reliance on color information, (3) expanding to multi-class classification using finer-grained safety scores, and (4) cross-city generalization studies with domain adaptation techniques.

References

[1] Dubey, A., et al. "Deep Learning the City: Quantifying Urban Perception at a Global Scale." ECCV 2016.

[2] Naik, N., et al. "Streetscore - Predicting the Perceived Safety of One Million Streetscapes." CVPR Workshop 2014.

[3] Dosovitskiy, A., et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." ICLR 2021.

[4] Zhang, F., et al. "Urban Perception: Employing Vision Transformers for Street-Level Image Analysis." IJGIS 2023.

[5] He, K., et al. "Deep Residual Learning for Image Recognition." CVPR 2016.

[6] MIT Media Lab. "Place Pulse 2.0 Dataset."
http://pulse.media.mit.edu/